

An empirical example: Four terminals and two areas

Daniel R. Miranda-Esquivel

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This is a worked example using a four terminals tree, distributed in two areas. Initially, all branches are equal.

```
# Four taxa and two areas
```

```
## Load libraries
```

```
library(blepd)
```

```
## Loading required package: ape
```

```
## Loading required package: picante
```

```
## Loading required package: vegan
```

```
## Loading required package: permute
```

```
##
```

```
## Attaching package: 'permute'
```

```
## The following object is masked from 'package:devtools':
```

```
##
```

```
##      check
```

```
## Loading required package: lattice
```

```
## This is vegan 2.6-6.1
```

```
## Loading required package: nlme
```

```
library(ggtree)
```

```
## ggtree v3.10.1 For help: https://yulab-smu.top/treedata-book/
```

```
##
```

```
## If you use the ggtree package suite in published research, please cite  
## the appropriate paper(s):
```

```
##
```

```
## Guangchuang Yu, David Smith, Huachen Zhu, Yi Guan, Tommy Tsan-Yuk Lam.
```

```
## ggtree: an R package for visualization and annotation of phylogenetic
```

```
## trees with their covariates and other associated data. Methods in
```

```
## Ecology and Evolution. 2017, 8(1):28-36. doi:10.1111/2041-210X.12628
```

```
##
```

```
## G Yu. Data Integration, Manipulation and Visualization of Phylogenetic
```

```
## Trees (1st ed.). Chapman and Hall/CRC. 2022. ISBN: 9781032233574
```

```
##
```

```
## S Xu, Z Dai, P Guo, X Fu, S Liu, L Zhou, W Tang, T Feng, M Chen, L
```

```
## Zhan, T Wu, E Hu, Y Jiang, X Bo, G Yu. ggtreeExtra: Compact
```

```

## visualization of richly annotated phylogenetic data. Molecular Biology
## and Evolution. 2021, 38(9):4039-4042. doi: 10.1093/molbev/msab166

##
## Attaching package: 'ggtree'

## The following object is masked from 'package:nlme':
##
##      collapse

## The following object is masked from 'package:ape':
##
##      rotate
library(ggplot2)

## Call the data

# Call the four terminals example

##data(package = "blepd")

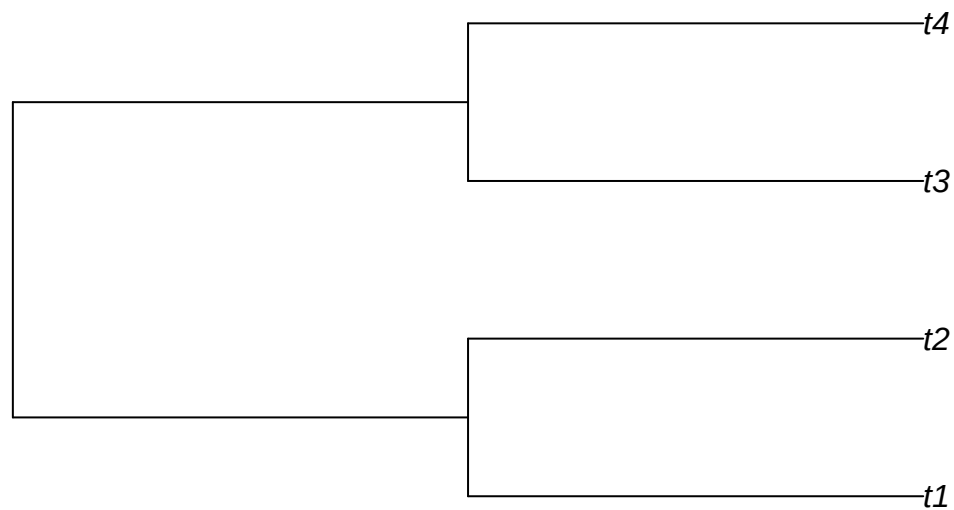
## trees

data(tree)

## Plot the tree

plot(tree,use.edge.length =T)

```



```
initialTree <- tree

## Load and plot the distribution

data(distribution)

##str(distribution)

dist4taxa <- distribution

## Change distribution to XY to be able to plot the data, otherwise, skip the next step

distXY <- matrix2XY(dist4taxa)

distXY

##   Terminal Area
## 1      t1    A1
## 2      t3    A1
## 3      t2    A2
## 4      t4    A2

## plotting

## the tree, using ggtree
```

```
plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle("Four terminals, all branches are == 1")

print(plotTree)
```

Four terminals, all branches are == 1

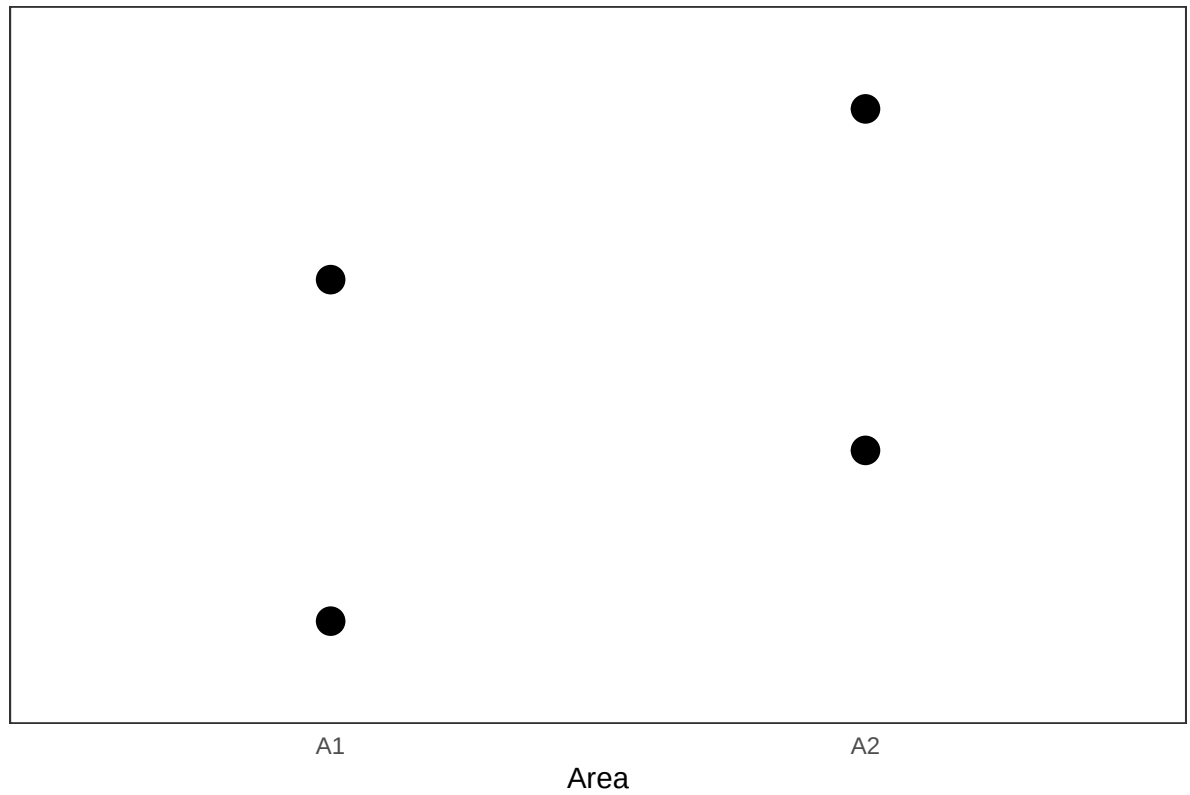


```
## The distribution using ggplot

plotDistrib <- ggplot(data=distXY,
                      aes(x= Area, y= Terminal , size =150)) +
  geom_point() +
  theme_bw() +
  theme( axis.text.y = element_blank(),
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        axis.line = element_line(colour = NA_character_),
        axis.ticks = element_blank(),
        legend.position = "none" ) +
  labs(title = "Distributions",
       y = "",
       x = "Area")
```

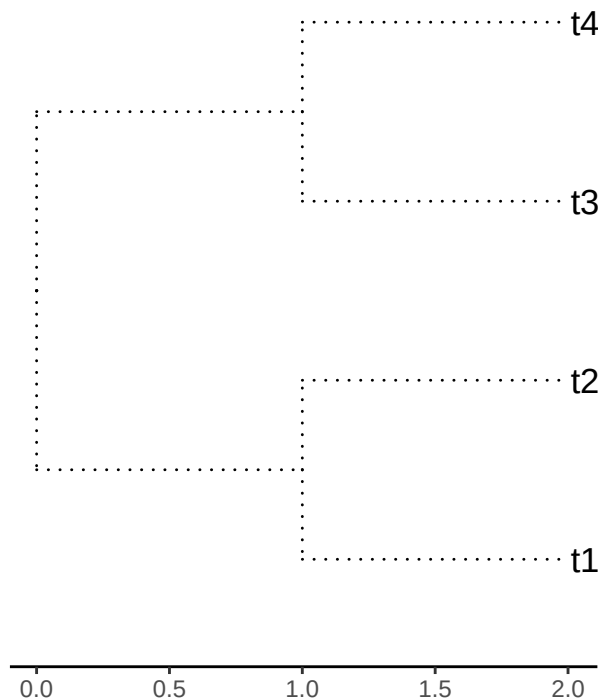
```
##  
print(plotDistrib)
```

Distributions

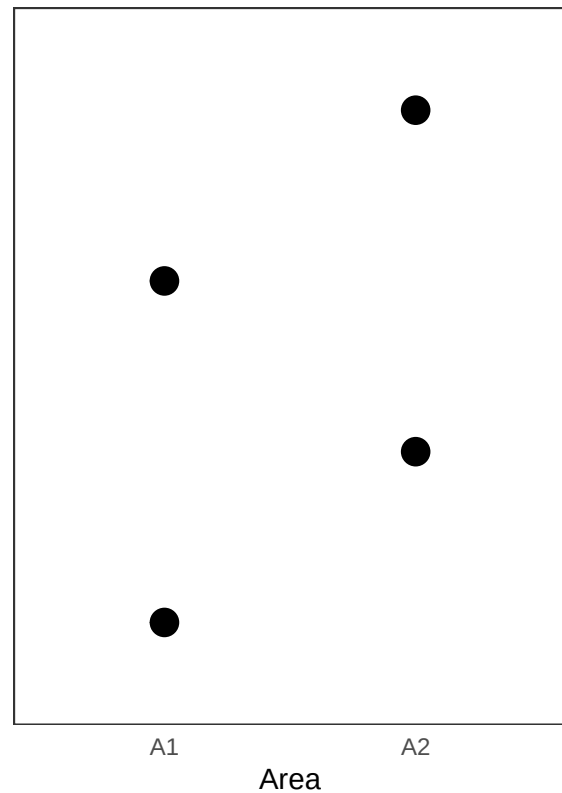


```
cowplot::plot_grid(plotTree, plotDistrib, ncol=2)
```

Four terminals, all branches are == 1



Distributions



```
# Check whether names in both objects, initialTree and dist4taxa, are the same.
```

```
##all(colnames(dist4taxa) == initialTree$tip.label)
```

```
all(colnames(dist4taxa) %in% initialTree$tip.label)
```

```
## [1] TRUE
```

```
#Report the branch length, and calculate the PD values.
```

```
initialTree$edge.length
```

```
## [1] 1 1 1 1 1 1
```

```
initialPD <- PDindex(tree=initialTree, distribution = dist4taxa)
```

```
initialPD
```

```
## [1] 4 4
```

```
#As expected there is a tie between both areas.
```

```
#Now, we can see the impact of swapping the branch lengths; we will use the three  
#available models to swap: "simpleswap","allswap","uniform", and we will swap the  
#three classes of branches: "terminals","internals","all".
```

```

##?swapBL

for( modelo in c("simplswap","allswap","uniform") ){

  for( rama in c("terminals","internals","all") ){

    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                  )
    printSwapBL(val, compact=TRUE)

    cat( "\n\n" )

  }

}

```

```

## model to test simplswap reps 100
##   A1A2
## 1  100
##
##
## model to test simplswap reps 100
##   A1A2
## 1  100
##
##
## model to test simplswap reps 100
##   A1A2
## 1  100
##
##
## model to test allswap reps 100
##   A1A2
## 1  100
##
##
## model to test allswap reps 100
##   A1A2
## 1  100
##
##
## model to test allswap reps 100
##   A1A2
## 1  100
##
##
## model to test uniform reps 100
##   A1A2

```

```
## 1 100
##
##
## model to test uniform reps 100
## A1A2
## 1 100
##
##
## model to test uniform reps 100
## A1A2
## 1 100
```

#As expected, as all braches are EQUAL nothing happens

#Now, let us see if the results could be related to the branch length (or not).

*## we could use a single command (*evalBranch*)*

##?evalBranch

```
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "terminals",
                           approach="all")
```

Loading required package: maps

##

Attaching package: 'phytools'

The following object is masked from 'package:vegan':

##

scores

The structure of the output

##str(AllTerminals)

Print it as a friendly output

```
printEvalBranch(AllTerminals, compact=FALSE)
```

```
##      node initialArea FinalArea Aproach %Delta
## 1 [t1]|1      A1A2      A2   lower -1.024
## 2 [t2]|2      A1A2      A1   lower -1.024
## 3 [t3]|3      A1A2      A2   lower -1.024
## 4 [t4]|4      A1A2      A1   lower -1.024
## 5 [t1]|1      A1A2      A1   upper  1.062
## 6 [t2]|2      A1A2      A2   upper  1.062
## 7 [t3]|3      A1A2      A1   upper  1.062
## 8 [t4]|4      A1A2      A2   upper  1.062
```



```
## Now the internals
```

```
AllInternals <- evalBranch(tree=initialTree,  
                           distribution = dist4taxa,  
                           branchToEval = "internals",  
                           approach="all")
```

```
printEvalBranch(AllInternals, compact=FALSE)
```

```
##           node initialArea FinalArea Aproach %Delta  
## 1 [node number:6:t1/t2] |6          A1A2      A1A2  lower      0  
## 2 [node number:7:t3/t4] |7          A1A2      A1A2  lower      0  
## 3 [node number:6:t1/t2] |6          A1A2      A1A2  upper      0  
## 4 [node number:7:t3/t4] |7          A1A2      A1A2  upper      0
```

```
#Now with a long branch
```

```
initialTree <- tree
```

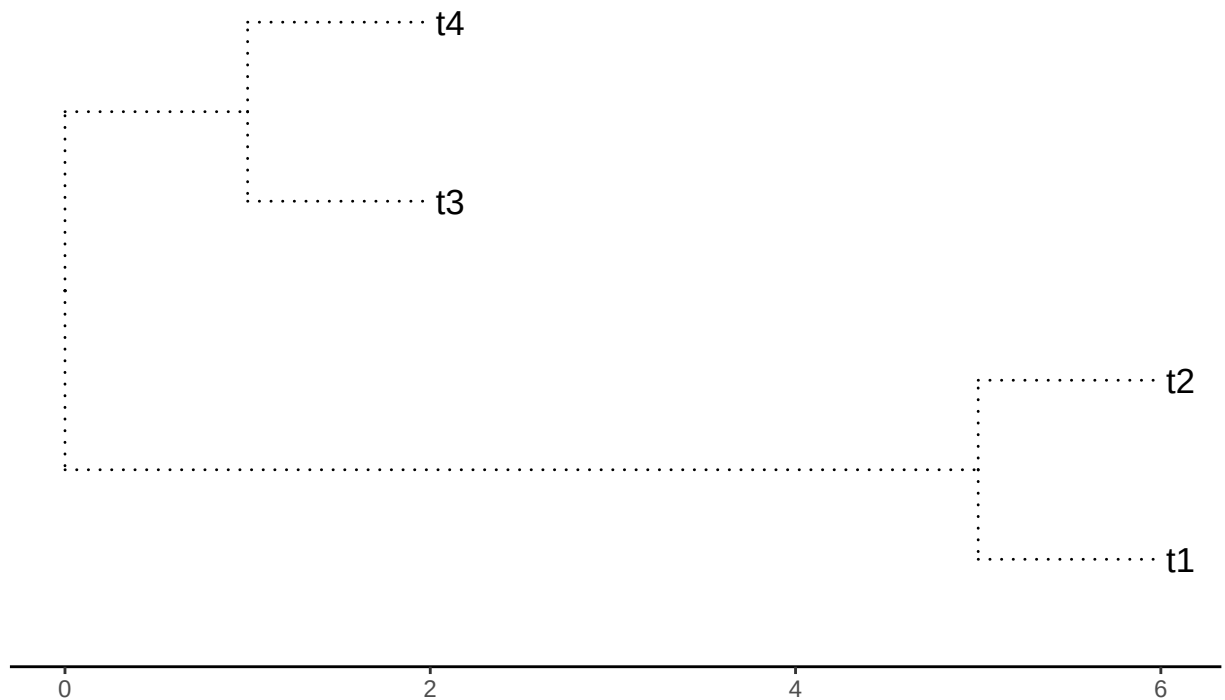
```
initialTree$edge.length[1] <- 5
```

```
plotTree <- ggtree(initialTree, ladderize=TRUE,  
                   color="black", size=0.5, linetype="dotted") +  
  geom_tiplab(size=4.5, color="black") +  
  theme_tree2() +  
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],  
                ". 2=",initialTree$edge.length[2],".",sep=""))
```

```
##
```

```
print(plotTree)
```

Four terminals. Branches 1=5. 2=1.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printSwapBL(val, compact=TRUE)

    cat( "\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
## A1A2
## 1 100
##
##
## model to test simpleswap reps 100
## A1A2
## 1 100
```

```

##
##
## model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 10   78 12
##
##
## model to test allswap reps 100
##   A1A2
## 1   100
##
##
## model to test allswap reps 100
##   A1A2
## 1   100
##
##
## model to test allswap reps 100
##   A1 A1A2 A2
## 1 34   39 27
##
##
## model to test uniform reps 100
##   A1A2
## 1   100
##
##
## model to test uniform reps 100
##   A1A2
## 1   100
##
##
## model to test uniform reps 100
##   A1 A2
## 1 50 50

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

##      node initialArea FinalArea Aproach %Delta
## 1 [t1]|1      A1A2      A2   lower -1.024
## 2 [t2]|2      A1A2      A1   lower -1.024
## 3 [t3]|3      A1A2      A2   lower -1.024
## 4 [t4]|4      A1A2      A1   lower -1.024
## 5 [t1]|1      A1A2      A1   upper  1.055
## 6 [t2]|2      A1A2      A2   upper  1.055
## 7 [t3]|3      A1A2      A1   upper  1.055
## 8 [t4]|4      A1A2      A2   upper  1.055

## Now the internals

```

```

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

##               node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2] |6          A1A2      A1A2  lower      0
## 2 [node number:7:t3/t4] |7          A1A2      A1A2  lower      0
## 3 [node number:6:t1/t2] |6          A1A2      A1A2  upper      0
## 4 [node number:7:t3/t4] |7          A1A2      A1A2  upper      0
# Now with a longer branch

initialTree <- tree

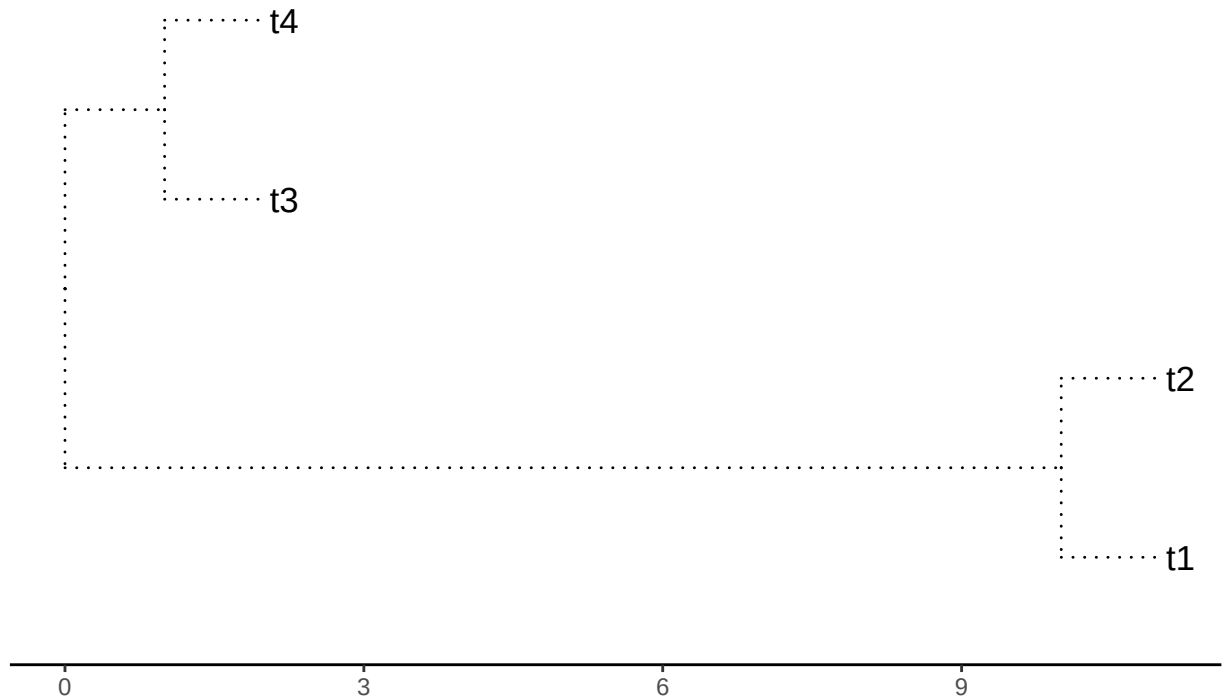
initialTree$edge.length[1] <- 10

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=10. 2=1.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printSwapBL(val, compact=TRUE)

    cat( "\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
## A1A2
## 1 100
##
##
## model to test simpleswap reps 100
## A1A2
## 1 100
```

```

##
##
## model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 18   63 19
##
##
## model to test allswap reps 100
##   A1A2
## 1   100
##
##
## model to test allswap reps 100
##   A1A2
## 1   100
##
##
## model to test allswap reps 100
##   A1 A1A2 A2
## 1 34   45 21
##
##
## model to test uniform reps 100
##   A1A2
## 1   100
##
##
## model to test uniform reps 100
##   A1A2
## 1   100
##
##
## model to test uniform reps 100
##   A1 A2
## 1 36 64

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

##      node initialArea FinalArea Aproach %Delta
## 1 [t1]|1      A1A2      A2   lower -1.024
## 2 [t2]|2      A1A2      A1   lower -1.024
## 3 [t3]|3      A1A2      A2   lower -1.024
## 4 [t4]|4      A1A2      A1   lower -1.024
## 5 [t1]|1      A1A2      A1   upper  1.022
## 6 [t2]|2      A1A2      A2   upper  1.022
## 7 [t3]|3      A1A2      A1   upper  1.022
## 8 [t4]|4      A1A2      A2   upper  1.022

## Now the internals

```

```

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

##           node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2] |6          A1A2      A1A2  lower      0
## 2 [node number:7:t3/t4] |7          A1A2      A1A2  lower      0
## 3 [node number:6:t1/t2] |6          A1A2      A1A2  upper      0
## 4 [node number:7:t3/t4] |7          A1A2      A1A2  upper      0
# With a short branch

initialTree <- tree

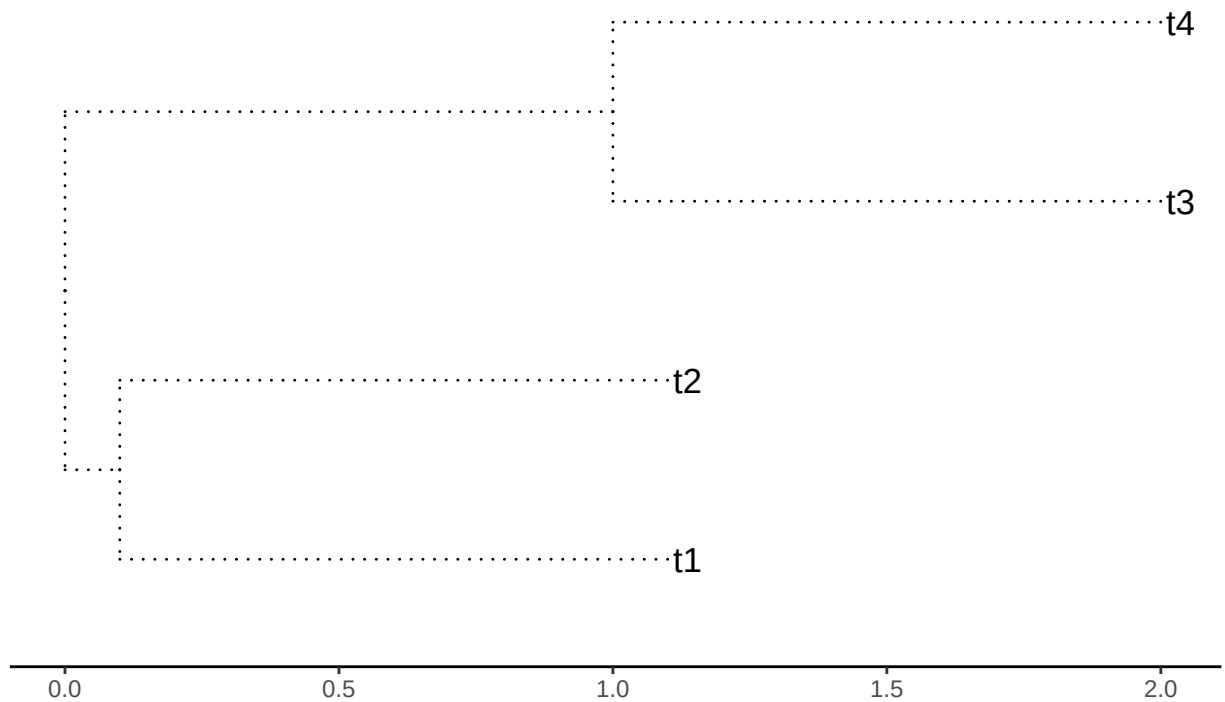
initialTree$edge.length[1] <- 0.1

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=0.1. 2=1.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printSwapBL(val, compact=TRUE)

    cat( "\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
## A1A2
## 1 100
##
##
## model to test simpleswap reps 100
## A1A2
## 1 100
```



```

##
##
## model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 13   77 10
##
##
## model to test allswap reps 100
##   A1A2
## 1   100
##
##
## model to test allswap reps 100
##   A1A2
## 1   100
##
##
## model to test allswap reps 100
##   A1 A1A2 A2
## 1 37   32 31
##
##
## model to test uniform reps 100
##   A1A2
## 1   100
##
##
## model to test uniform reps 100
##   A1A2
## 1   100
##
##
## model to test uniform reps 100
##   A1 A2
## 1 53 47

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

##      node initialArea FinalArea Aproach %Delta
## 1 [t1]|1      A1A2      A2   lower -1.024
## 2 [t2]|2      A1A2      A1   lower -1.024
## 3 [t3]|3      A1A2      A2   lower -1.024
## 4 [t4]|4      A1A2      A1   lower -1.024
## 5 [t1]|1      A1A2      A1   upper  1.051
## 6 [t2]|2      A1A2      A2   upper  1.051
## 7 [t3]|3      A1A2      A1   upper  1.051
## 8 [t4]|4      A1A2      A2   upper  1.051

## Now the internals

```

```

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

##               node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2] |6          A1A2      A1A2  lower      0
## 2 [node number:7:t3/t4] |7          A1A2      A1A2  lower      0
## 3 [node number:6:t1/t2] |6          A1A2      A1A2  upper      0
## 4 [node number:7:t3/t4] |7          A1A2      A1A2  upper      0
##With a long branch in brach 2

initialTree <- tree

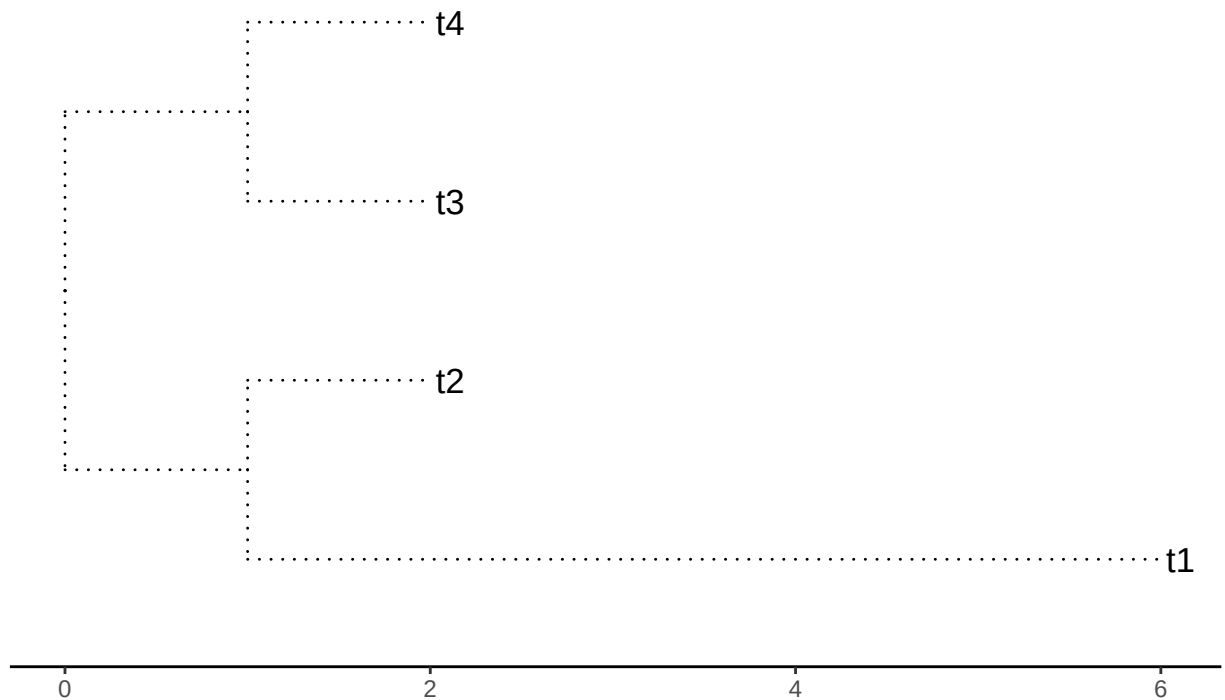
initialTree$edge.length[2] <- 5

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2], ".", sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=1. 2=5.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printSwapBL(val, compact=TRUE)

    cat( "\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
##   A1 A2
## 1 68 32
##
##
## model to test simpleswap reps 100
##   A1
## 1 100
```

```

##
##
## model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 73   13 14
##
##
## model to test allswap reps 100
##   A1 A2
## 1 50 50
##
##
## model to test allswap reps 100
##   A1
## 1 100
##
##
## model to test allswap reps 100
##   A1 A1A2 A2
## 1 32   34 34
##
##
## model to test uniform reps 100
##   A1 A2
## 1 52 48
##
##
## model to test uniform reps 100
##   A1
## 1 100
##
##
## model to test uniform reps 100
##   A1 A2
## 1 52 48

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

##      node initialArea FinalArea Aproach %Delta
## 1 [t1]|1          A1          A2  lower -80.193
## 2 [t2]|2          A1          A1  lower      0
## 3 [t3]|3          A1          A1  lower      0
## 4 [t4]|4          A1          A1  lower      0
## 5 [t1]|1          A1          A1  upper      0
## 6 [t2]|2          A1          A2  upper 400.005
## 7 [t3]|3          A1          A1  upper      0
## 8 [t4]|4          A1          A2  upper 400.005

## Now the internals

```

```

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

##               node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2] |6          A1          A1  lower      0
## 2 [node number:7:t3/t4] |7          A1          A1  lower      0
## 3 [node number:6:t1/t2] |6          A1          A1  upper      0
## 4 [node number:7:t3/t4] |7          A1          A1  upper      0
# Now with a longer branch2

initialTree <- tree

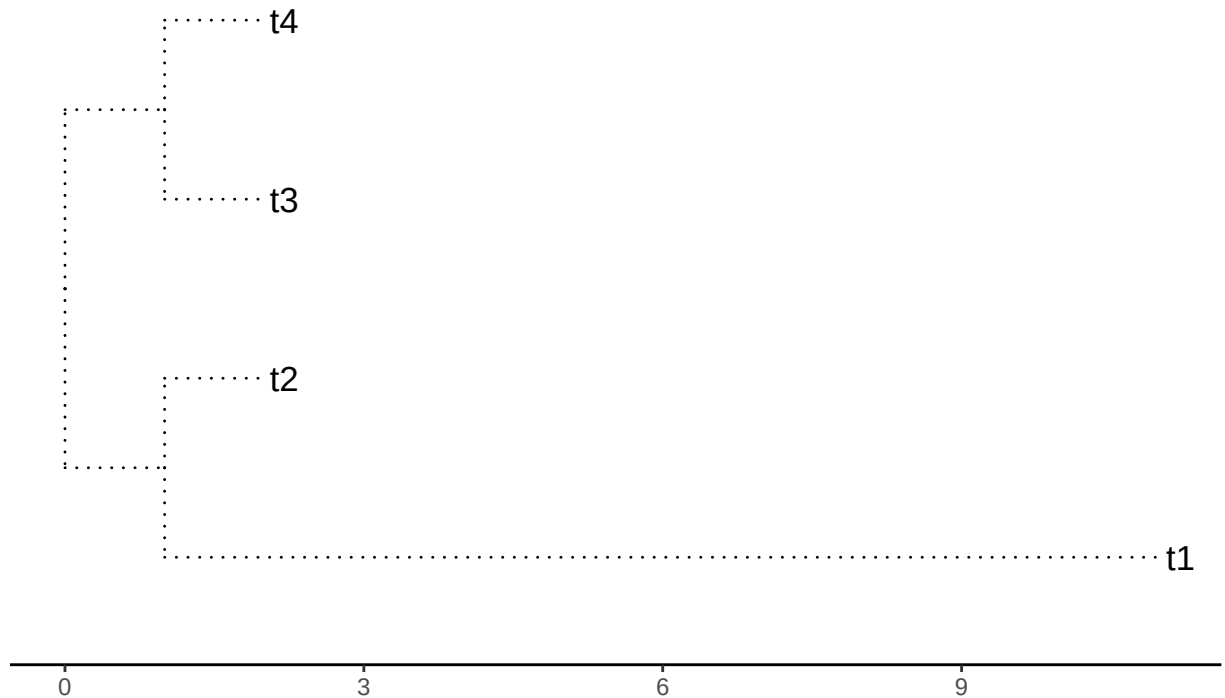
initialTree$edge.length[2] <- 10

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=1. 2=10.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printSwapBL(val, compact=TRUE)

    cat( "\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
##   A1 A2
## 1 74 26
##
##
## model to test simpleswap reps 100
##   A1
## 1 100
```

```

##
##
## model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 83   10  7
##
##
## model to test allswap reps 100
##   A1 A2
## 1 41 59
##
##
## model to test allswap reps 100
##   A1
## 1 100
##
##
## model to test allswap reps 100
##   A1 A1A2 A2
## 1 37   32 31
##
##
## model to test uniform reps 100
##   A1 A2
## 1 50 50
##
##
## model to test uniform reps 100
##   A1
## 1 100
##
##
## model to test uniform reps 100
##   A1 A2
## 1 54 46

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

##      node initialArea FinalArea Aproach %Delta
## 1 [t1]|1          A1          A2  lower -90.096
## 2 [t2]|2          A1          A1  lower      0
## 3 [t3]|3          A1          A1  lower      0
## 4 [t4]|4          A1          A1  lower      0
## 5 [t1]|1          A1          A1  upper      0
## 6 [t2]|2          A1          A2  upper 900.004
## 7 [t3]|3          A1          A1  upper      0
## 8 [t4]|4          A1          A2  upper 900.004

## Now the internals

```

```

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

##               node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2] |6          A1          A1  lower      0
## 2 [node number:7:t3/t4] |7          A1          A1  lower      0
## 3 [node number:6:t1/t2] |6          A1          A1  upper      0
## 4 [node number:7:t3/t4] |7          A1          A1  upper      0
#now with a longer branch

initialTree <- tree

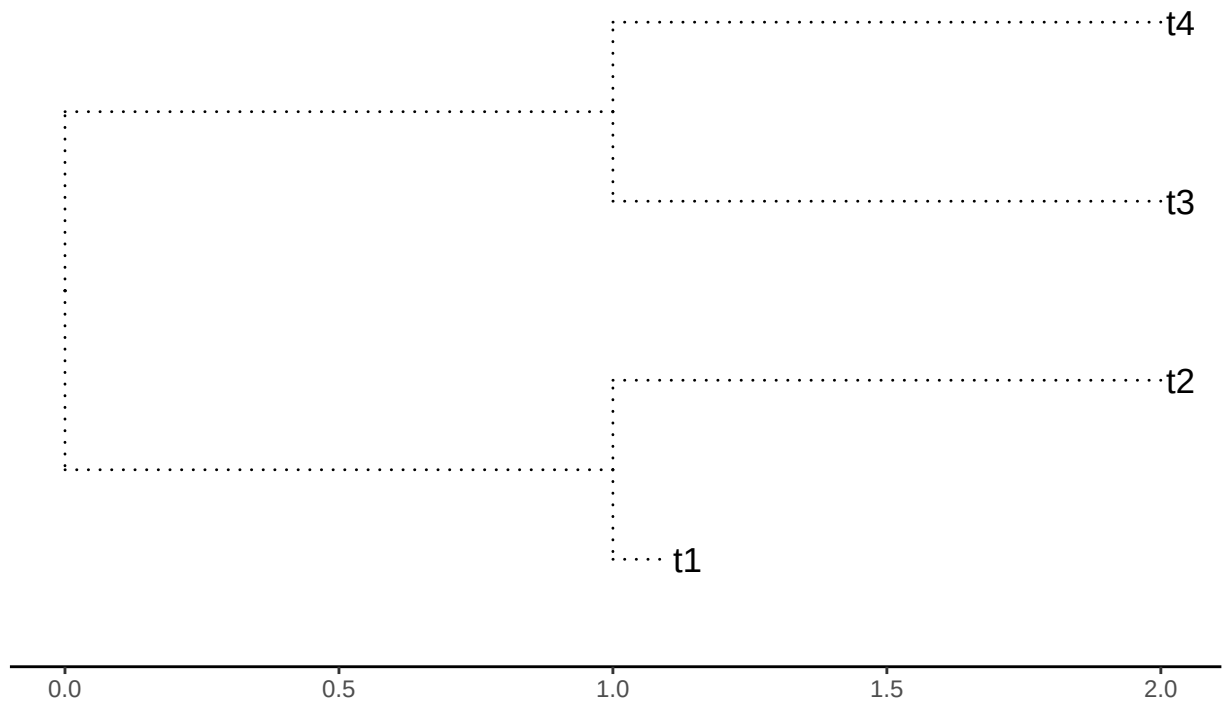
initialTree$edge.length[2] <- 0.1

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)

```


Four terminals. Branches 1=1. 2=0.1.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printSwapBL(val, compact=TRUE)

    cat( "\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
##   A1 A2
## 1 30 70
##
##
## model to test simpleswap reps 100
##   A2
## 1 100
```

```

##
##
## model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 18   11 71
##
##
## model to test allswap reps 100
##   A1 A2
## 1 42 58
##
##
## model to test allswap reps 100
##   A2
## 1 100
##
##
## model to test allswap reps 100
##   A1 A1A2 A2
## 1 35   40 25
##
##
## model to test uniform reps 100
##   A1 A2
## 1 56 44
##
##
## model to test uniform reps 100
##   A2
## 1 100
##
##
## model to test uniform reps 100
##   A1 A2
## 1 45 55

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

##      node initialArea FinalArea Aproach %Delta
## 1 [t1]|1          A2          A2  lower      0
## 2 [t2]|2          A2          A1  lower -90.115
## 3 [t3]|3          A2          A2  lower      0
## 4 [t4]|4          A2          A1  lower -90.115
## 5 [t1]|1          A2          A1  upper 900.408
## 6 [t2]|2          A2          A2  upper      0
## 7 [t3]|3          A2          A1  upper  90.005
## 8 [t4]|4          A2          A2  upper      0

## Now the internals

```

```
AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "internals",
                           approach="all")
```

```
printEvalBranch(AllInternals, compact=FALSE)
```

```
##           node initialArea FinalArea Approach %Delta
## 1 [node number:6:t1/t2] |6          A2          A2  lower      0
## 2 [node number:7:t3/t4] |7          A2          A2  lower      0
## 3 [node number:6:t1/t2] |6          A2          A2  upper      0
## 4 [node number:7:t3/t4] |7          A2          A2  upper      0
```